

PS2 - Smart Municipal Complaint & Resolution Intelligence Platform

Complete Product Requirements & Technical Specification (Solapur Edition)

Version: 2.0 (Finalized) **Date:** September 2026 **Project Type:** Buildathon / Hackathon **Pilot City:** Solapur, Maharashtra, India

1. Executive Summary

The Problem: Municipal authorities receive complaints about potholes, garbage, drainage, water leakage, damaged streetlights, and other civic issues through multiple channels. Complaints are misrouted, duplicates go unnoticed, urgent issues get buried, and officers lack a consolidated view of recurring problems.

Real-World Validation: In July 2026, Solapur Municipal Corporation (SMC) took strict action against **51 officers and employees** for delaying online complaint resolution. **21 employees were fined ₹4,000** for not resolving complaints on the "My Solapur" app, and **23 officers/employees were fined ₹25,900** for delaying applications under the Maharashtra Right to Public Services Act. The Municipal Commissioner issued a strict warning: *"No further delays will be tolerated."*

Our Solution: A dual-platform (Citizen Mobile + Officer Web) intelligence system that uses Computer Vision to classify complaint images, Natural Language Processing to detect duplicates, Geospatial analysis for hotspot detection, and Generative AI to answer operational questions. The system actively detects emerging crises, groups duplicate complaints, and verifies resolutions using before/after image comparison. **We use real Solapur ward data, real department structures, and real geography—only officer/worker names are synthetic.**

Key Innovation: We transform individual complaints into actionable operational intelligence by: 1. **Auto-classifying** complaints from citizen photos (Pothole vs Garbage vs Drainage) 2. **Grouping duplicates** using text similarity + location proximity 3. **Detecting emerging crises** through statistical spike detection 4. **Verifying resolutions** using automated image comparison 5. **Balancing workloads** across departments using real-time analytics 6. **Automating SLA enforcement** to prevent the exact delays that got 51 SMC officers fined

2. Core Problem Statement (From PS2)

Build a Smart Municipal Complaint & Resolution Intelligence Platform that enables citizens and municipal staff to report, process, prioritize, assign, track, and resolve civic complaints using a combination of web/mobile application development, backend systems, AI/ML, geospatial analysis, and Generative AI.

Key Intelligence Objectives:

- What problems are currently being reported?
- Where are they occurring?
- Which complaints are urgent?
- Are multiple complaints referring to the same underlying issue?
- Which departments should handle them?
- Which issues are repeatedly occurring?
- Which areas are experiencing an increasing number of complaints?
- Which complaints require immediate attention?

3. Pilot City: Solapur, Maharashtra

3.1 Why Solapur?

Reason	Details
Real Problem Validation	51 SMC officers were fined in July 2026 for delaying complaint resolution. Our platform directly solves this.
Real Administrative Structure	SMC has 8 zones, 26 election wards, 98 census wards, and 102 councillors.
Real Department Structure	Solid Waste Management, Public Health, Roads & Infrastructure, Water Supply, Street Lighting, Urban Planning.
Real Geography	OpenStreetMap provides detailed Solapur maps with ward boundaries.
Real Population Data	Each ward has official population, SC/ST population, and boundary descriptions.

3.2 Solapur's Administrative Structure

Structure	Details
City	Solapur, Maharashtra
Zones	8 administrative zones
Election Wards	26 wards
Census Wards	98 wards (for census purposes)
Councillors	102
Key Departments	Urban Planning, Public Health, Sanitation, Waste Management, Street Lighting, Water Supply, Roads & Infrastructure

3.3 Solapur Ward Data (Real)

Example: Ward 1

- **Name:** Kasbe Solapur-1 Shelgi (Gaonthan)
- **Population:** 37,955 (SC: 10,319, ST: 1,271)
- **Boundaries:** Karamba Road, Ganesh Nagar, Tuljapur Naka, Javalkar Wasti, Ramabai Ambedkar Nagar, Chandak High School, Hanuman Nagar, Vardhman Nagar, Rupa Bhawan Temple, Maddi Wasti, State Bank Colony, Shelgi Gaonthan Area.
- **Councillors:** 4

All 26 wards will be pre-seeded in the database with their real names, populations, and approximate geographical boundaries.

3.4 Real Departments (From SMC)

Department	Primary Responsibilities
Solid Waste Management	Garbage collection, waste disposal, cleanliness
Public Health & Sanitation	Drainage, water leakage, public health hazards, disease prevention
Roads & Infrastructure	Potholes, road obstructions, street maintenance, bridges
Street Lighting	Damaged streetlights, electrical issues

Department	Primary Responsibilities
Water Supply	Water leakage, supply issues, pipeline maintenance
Urban Planning	Property-related complaints, construction violations, zoning
Public Works	General infrastructure, building maintenance

4. Target Users & Dashboard Strategy

We have defined **4 distinct dashboards** to cover all stakeholders:

Dashboard	Platform	Primary User	Primary Goal
1. Citizen Portal	Mobile + Web App	Citizens of Solapur	Submit complaints, track status, view nearby issues on a public map of Solapur.
2. Command Center	Web App	Municipal Administrator, Department Head	City-wide view, heatmaps by ward, emerging crisis alerts, department workload monitoring.
3. Operations Desk	Web App	Municipal Officer, Department Admin	Review AI classifications, assign/reassign complaints, update status, manage duplicate groups.
4. Field Worker	Mobile App	Field Workers (sanitation, repair crews)	Receive assignments, update GPS location, upload resolution photos, mark tasks complete.

5. Finalized Feature Scope (MVP)

Core Features (Must Have)

Module	Feature	Description
Complaint Submission	Citizen Portal	Text + Photo + GPS + Category selection (or AI auto-detect). Auto-detects Solapur ward from GPS.
Complaint Lifecycle	Workflow	Submitted → Under Review → Assigned → In Progress → Resolution Submitted → Verified → Closed.
AI - Image Classification	Auto-Detect Category	Use a pre-trained model (ResNet50) to classify images as Pothole, Garbage, Drainage, Streetlight, Water Leakage, Other.

Module	Feature	Description
AI - Text Classification	Auto-Tag Severity	Use NLP to extract keywords and assign severity (Low/Medium/High/Critical) based on urgency words (e.g., "danger", "flooded", "collapsed").
AI - Department Suggestion	Auto-Routing	Suggest the correct SMC department based on category + location (e.g., "Garbage" → Solid Waste Management).
AI - Duplicate Detection	Group Similar Complaints	Compare text similarity (TF-IDF) + location proximity (within 100m). Show "Related Complaints" on the detail page.
AI - Trend Detection	Emerging Crisis Alert	Detect sudden spikes in complaints (e.g., "Drainage complaints in Ward 4 increased 240% in 24h"). Flag on dashboard.
Prioritization	Rule-Based Scoring	Severity + Duration + Duplicates + Safety Risk → Priority Score (Critical → Low).
Resolution Workflow	Field Worker App	Receive tasks, update status (In Progress → Resolved), upload Before/After photos.
GenAI Assistant	Officer Chat	Answers questions using actual complaint data (e.g., "What are major unresolved problems in Ward 4?").
Dashboard	Heatmap & Charts	Show open complaints by category, by ward (1-26), overdue complaints, department workload.
SLA Enforcement	Auto-Escalation	Automatically escalate complaints that exceed SLA deadlines (preventing the delays that got 51 SMC officers fined).

Stretch Features (Innovation)

Module	Feature	Description
Resolution Verification	Before/After Comparison	Use image similarity (Structural Similarity Index) to automatically check if "After" photo is actually the same location and appears fixed.
Public Map	Citizen View	Show open complaints on a map of Solapur. Allow filtering by ward and category. Read-only, no login required.

Module	Feature	Description
SLA Prediction	Overdue Alert	Predict if a complaint is likely to miss its SLA deadline based on department workload.
Department Workload	Resource Recommendation	Suggest reallocating resources to overloaded departments (e.g., "Solid Waste has 25 open complaints in Ward 3. Recommend transferring 2 officers from Public Works.").
Auto-Assignment	Smart Routing	Automatically assign complaints to specific field workers based on location and availability.
Trust Score	Citizen Reputation	Citizens earn trust scores for verified complaints. High trust = faster processing priority.

6. Anti-Failure Architecture (Preventing the 51-Officer Problem)

We build **five layers of accountability** to prevent the exact delays that got 51 SMC officers fined.

6.1 Problem: Worker Uploads False "After" Image

Solution	How It Works
SSIM Automated Verification	Compares "Before" and "After" photos pixel-by-pixel. Low SSIM score (< 0.3) = flag as "Suspicious."
GPS Matching	Compare GPS of "After" photo with original complaint GPS. > 50m difference = flag.
Timestamp Check	Compare timestamp of "After" photo with assignment time. > 48 hours = flag.
Manual Random Audit	10% of resolved complaints randomly selected for supervisor review (deters cheating).

6.2 Problem: Complaint Never Assigned

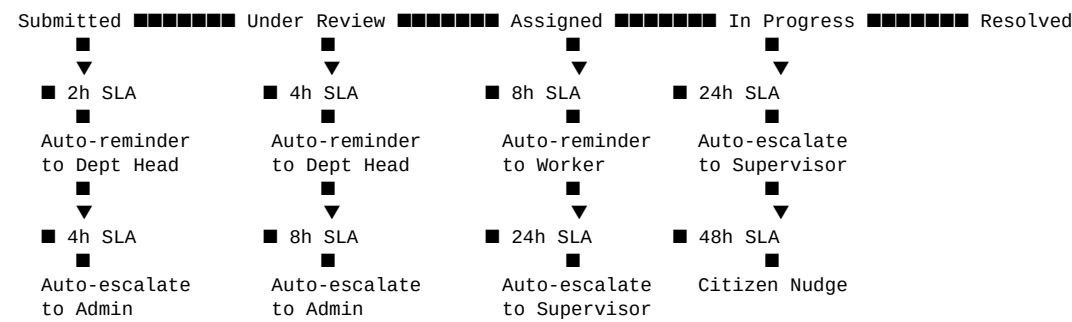
Solution	How It Works
SLA Timer - Submitted	2 hours: Auto-reminder to Department Head. 4 hours: Auto-escalate to Municipal Administrator.
SLA Timer - Under Review	4 hours: Auto-reminder to Department Head. 8 hours: Auto-escalate to Municipal Administrator.
Visual Dashboard	"Pending Assignment" count with red clock icon.

6.3 Problem: Complaint Assigned But Never Solved

Solution	How It Works
SLA Timer - In Progress	8 hours: Auto-reminder to worker. 24 hours: Auto-escalate to supervisor.
Supervisor Override	Department Head can re-assign, escalate, or mark as "Needs Support."

Solution	How It Works
Citizen Nudge	After 48 hours, auto-send notification to citizen: "We're still working on your complaint."
"Stalled" Widget	Dashboard shows all complaints stuck "In Progress" for over 24 hours.

6.4 SLA Escalation Workflow



6.5 Real-World Validation in Demo

"In July 2026, 51 SMC officers were fined for delaying complaint resolution. Our platform prevents this. If a complaint isn't assigned within 2 hours, the Department Head gets a reminder. If it's still unassigned after 4 hours, it escalates to the Municipal Administrator. If a worker delays, the system sends reminders every 8 hours. We're not just building another complaint app—we're building accountability."

7. Detailed User Workflows (Solapur Context)

Workflow A: Citizen Reports Garbage (Ward 1)

Step 1: Citizen Submission

Time: 9:00 AM **Location:** Kasbe Solapur-1 Shelgi (Ward 1), near Tuljapur Naka **Device:** Citizen's Mobile Phone

Citizen Amit's Actions: 1. Opens the **Citizen Portal** app. 2. Takes a photo of overflowing garbage near Tuljapur Naka. 3. Types: "Garbage has not been collected for 4 days near Tuljapur Naka. It's attracting stray animals and smells terrible." 4. The app auto-captures GPS: (17.6599, 75.9064). 5. Auto-detects **Ward 1** from GPS. 6. Clicks "Submit Complaint".

What Happens in Backend:

Step	Process	Result
1	Image Classification (ResNet50)	Detected: "Garbage / Waste" (Confidence: 92%)
2	Text Classification (TF-IDF)	Keywords: "garbage", "not collected", "4 days" → Severity: High
3	Department Suggestion	Category: Garbage → Solid Waste Management
4	Duplicate Detection	Checks similar text within 300m in last 7 days. Finds 3 related complaints. Creates "Related Complaints Group ID: G-12."

Step	Process	Result
5	Prioritization Score	Severity: High (40) + Duration: 4 days (20) + Related complaints: 3 (15) + Safety Risk: Public Health Hazard (25) = Total: 100/100 → Priority: CRITICAL
6	Ward Detection	GPS maps to Ward 1 (Kasbe Solapur-1 Shelgi)

Data Saved in Database:

```
{
  "id": "COMP-2026-001",
  "citizen_id": "CIT-AMIT-001",
  "title": "Garbage not collected for 4 days",
  "description": "Garbage has not been collected for 4 days near Tuljapur Naka. It's attracting stray animals and smells terrible.",
  "category": "garbage",
  "category_confidence": 0.92,
  "severity": "high",
  "priority_score": 100,
  "priority_label": "critical",
  "suggested_department": "solid_waste",
  "related_complaint_group_id": "G-12",
  "ward_id": 1,
  "ward_name": "Kasbe Solapur-1 Shelgi",
  "location_lat": 17.6599,
  "location_lng": 75.9064,
  "image_url": "cloudinary.com/complaints/COMP-2026-001.jpg",
  "status": "submitted",
  "created_at": "2026-09-08 09:00:00"
}
```

Step 2: Operations Desk Review

Time: 9:15 AM **Device:** Municipal Officer Priya's Laptop (Operations Desk)

What Priya Sees:

New Complaints Awaiting Review (12)

COMP-2026-001: Garbage not collected for 4 days

Category: Garbage (AI: 92% confidence)

Severity: HIGH

Priority: CRITICAL (Score: 100/100)

Suggested Dept: Solid Waste Management

Ward: Ward 1 (Kasbe Solapur-1 Shelgi)

Related Complaints: 3 found

- COMP-2025-450: Overflowing trash (2 days ago)
- COMP-2025-423: Garbage pile-up (3 days ago)
- COMP-2025-399: Waste not collected (5 days ago)

EMERGING PATTERN DETECTED:

- Garbage complaints in Ward 1 increased by 240% compared to last week.

Approve & Assign

Edit

Merge Duplicates

Priya's Actions: 1. Reviews the AI classification. 2. Confirms category: **Garbage** (correct). 3. Clicks **"Approve & Assign"**. 4. Selects department: **Solid Waste Management**. 5. Assigns to Field Worker: **Ramesh - Zone 1**. 6. Adds note: *"This is urgent. 4 days overdue. Please prioritize. Ward 1 has seen a 240% spike in garbage complaints."*

Step 3: Field Worker Assignment

Time: 9:20 AM **Device:** Field Worker Ramesh's Mobile Phone

Push Notification to Ramesh:

■ CRITICAL ASSIGNMENT

Complaint: Garbage not collected for 4 days
Location: Tuljapur Naka, Ward 1 (Kasbe Solapur-1 Shelgi)
Priority: CRITICAL
Note: 4 days overdue. Please prioritize.

Ramesh's Actions: 1. Opens the **Field Worker** app. 2. Sees the assignment with GPS location. 3. Navigates to Tuljapur Naka. 4. Takes a photo of the garbage pile. 5. Marks status: **"In Progress"**. 6. Adds note: *"Arrived on site. Garbage pile is large. Requesting additional support."*

Data Saved:

```
{
  "complaint_id": "COMP-2026-001",
  "status": "in_progress",
  "assigned_to": "USER-RAMESH-001",
  "field_photo": "cloudinary.com/field-photos/COMP-2026-001-progress.jpg",
  "updated_at": "2026-09-08 09:45:00",
  "note": "Arrived on site. Garbage pile is large. Requesting additional support."
}
```

Step 4: Resolution

Time: 11:00 AM **Device:** Ramesh's Mobile Phone

Ramesh's Actions: 1. Clears the garbage with his team. 2. Takes a **"After"** photo of the clean area. 3. Clicks **"Mark Resolved"**. 4. Adds note: *"Garbage cleared. Area sanitized. Tuljapur Naka is clean."*

Backend Processing: 1. Before/After Verification:

- Compares original photo (submitted by citizen) with the "After" photo.
- Uses Structural Similarity Index (SSIM) to compare images.
- SSIM Score: **0.12** (very low → indicates significant change → likely fixed).
- System suggests: **■ Likely Resolved**.

2. Status Update:

- Complaint status: **"Resolution Submitted"**.

Data Saved:

```
{
  "complaint_id": "COMP-2026-001",
  "status": "resolution_submitted",
  "after_photo_url": "cloudinary.com/resolutions/COMP-2026-001-after.jpg",
  "ssim_score": 0.12,
  "verification_suggestion": "likely_resolved",
  "resolved_at": "2026-09-08 11:00:00",
  "note": "Garbage cleared. Area sanitized. Tuljapur Naka is clean."
}
```

Step 5: Citizen Verification

Time: 11:30 AM **Device:** Citizen Amit's Mobile Phone

Notification to Amit:

■ Complaint COMP-2026-001 marked as Resolved.
Please verify: Is the issue fixed?
[■ Yes, it's fixed] [■ No, it's not fixed]

Amit's Action:

- Clicks **■ Yes, it's fixed**.

Data Saved:

```
{
  "complaint_id": "COMP-2026-001",
  "citizen_verification": "confirmed",
  "status": "closed",
  "closed_at": "2026-09-08 11:35:00"
}
```

Step 6: GenAI Summary

Time: 12:00 PM **Device:** Municipal Administrator's Laptop

Administrator's Query:

""Summarize waste management trends in Ward 1 this week.""

GenAI Assistant Response:

- Waste Management Trends - Ward 1 (Kasbe Solapur-1 Shelgi)
Week 36 (Sep 1-7, 2026)
- Total Garbage Complaints: 23 (▲ 240% vs last week)
- Critical Complaints: 4
 - COMP-2026-001: Tuljapur Naka (Resolved)
 - COMP-2026-003: Ganesh Nagar (In Progress)
 - COMP-2026-005: Karamba Road (Pending)
 - COMP-2026-007: Hanuman Nagar (Pending)
- Resolved: 15 (65% resolution rate)
 - Average resolution time: 2.1 days (SLA: 3 days)
- Pending: 8
 - Average pending duration: 3.4 days
 - Oldest pending: COMP-2026-008 (Day 5)
- Emerging Issue: Ward 1
 - Garbage complaints increased by 240%.
 - Suspected cause: Staff shortage (2 workers on leave).
 - Recommendation: Deploy additional workers to Ward 1 for next 2 weeks.
- Recommendations:
 1. Deploy 2 additional workers to Ward 1 immediately.
 2. Schedule a cleanliness drive in Ganesh Nagar and Karamba Road.
 3. Send automated reminders for 4 pending complaints due today.

Workflow B: Pothole with Duplicate Detection (Ward 4)

Step 1: Citizen A Reports

Time: 10:00 AM **Citizen A:** "Huge pothole outside City Mall." **GPS:** (17.6700, 75.9100) → Auto-detects Ward 4

Step 2: Citizen B Reports (2 Hours Later)

Time: 12:00 PM **Citizen B:** "Deep pothole near City Mall entrance." **GPS:** (17.6702, 75.9103) → Auto-detects Ward 4

Step 3: AI Duplicate Detection

Backend Processing: 1. **Text Similarity** (TF-IDF):

- "Huge pothole outside City Mall" vs "Deep pothole near City Mall entrance"
- Similarity Score: **0.82** (highly similar)

2. **Location Proximity:**

- Distance between GPS points: **~35 meters**
- Threshold: 100 meters → ■ **Within range**

3. **Ward Matching:**

- Both complaints detected in **Ward 4**.

4. **Duplicate Detection Result:**

- System detects both complaints refer to the same pothole.
- Creates "**Duplicate Group ID: P-05**".
- Both complaints are linked in the database.

Officer Dashboard:

Layer	Technology	Rationale
Mobile App (Citizen + Field Worker)	React Native (Expo)	One codebase for both. Camera + GPS support.
Web App (Command Center + Operations)	Next.js (React)	Server-side rendering for charts, maps, and dashboards.
Backend API	FastAPI (Python)	Unified server for AI (Image Classification + NLP) and APIs.
Database	PostgreSQL + PostGIS + pgvector	PostGIS for geospatial queries (ward boundaries, radius). Pgvector for RAG embeddings.
Mapping	Leaflet with OpenStreetMap	Free and open-source. Solapur map available.
AI - Image Classification	Pre-trained ResNet50 (or MobileNet for speed)	Fine-tune on a small dataset or use zero-shot classification. Detects Potholes, Garbage, Drainage, etc.
AI - Text Classification	scikit-learn (TF-IDF) or zero-shot (Hugging Face)	Lightweight, fast. No training required.
AI - Duplicate Detection	TF-IDF + Cosine Similarity + Geospatial (PostGIS)	Combines text similarity AND location proximity.
AI - Trend Detection	Simple Statistical Anomaly (Z-score)	Compares last 24h complaints against 7-day average. If > 2 standard deviations, flag as "Emerging."
AI - Before/After	Structural Similarity Index (SSIM)	Compares two photos pixel-by-pixel. If they are too different, the issue is likely fixed.
GenAI (RAG)	OpenAI GPT-3.5 + pgvector	Convert complaints to vectors. Retrieve relevant complaints for context.
Authentication	JWT	Stateless, works for mobile and web.

10. Database Schema (Conceptual - Key Tables)

Table	Description	Solapur-Specific Columns
users	Citizens, Officers, Admins, Field Workers. Role-based.	role, department, zone_id
wards	Solapur's 26 real wards.	ward_id, ward_name, zone_id, population, sc_population, st_population, boundary_geojson
zones	Solapur's 8 administrative zones.	zone_id, zone_name, jurisdiction_description

Table	Description	Solapur-Specific Columns
complaints	Core table.	ward_id, zone_id, category (AI predicted), severity (AI predicted), priority (rule-based), status, location_lat, location_lng, assigned_department, assigned_to, related_complaint_group_id, sla_reminder_sent_at, escalated_at
images	Stores Cloudinary URLs for complaint photos.	image_url, image_type (before/after), ssim_score
ai_insights	Stores classification results.	category, category_confidence, severity, severity_confidence, suggested_department
duplicate_groups	Groups related complaints together.	group_id, complaint_ids, ward_id, created_at
emerging_alerts	Stores detected spikes.	ward_id, category, spike_percentage, alert_level, created_at
complaint_embeddings	Vector embeddings for RAG (pgvector).	complaint_id, embedding
audit_logs	Tracks all status changes and officer actions.	user_id, action, complaint_id, timestamp
sla_audit	Tracks SLA violations and escalations.	complaint_id, sla_type, expected_time, actual_time, action_taken

11. GenAI Assistant - Query Examples (Solapur Context)

User	Question	System Response
Administrator	"What are the major unresolved problems in Ward 1?"	"5 unresolved complaints in Ward 1 (Kasbe Solapur-1 Shelgi): 3 Garbage, 1 Pothole, 1 Drainage. Average resolution time: 4.2 days."
Officer	"Why has this complaint been marked high priority?"	"Severity: High (Duration: 3 days). Related complaints: 2 within 200m. Safety Risk: Public health hazard. Ward: 1 (Kasbe Solapur-1 Shelgi)."
Administrator	"Which wards have repeatedly reported drainage problems?"	"Ward 4 (City Center): 12 drainage complaints in last month (▲140% vs previous month). Ward 7 (Railway Colony): 8 drainage complaints. Recommend inspecting main drain pipeline in Ward 4."

User	Question	System Response
Department Head	"Generate a daily operational summary for Zone 3."	"Zone 3: 52 complaints received, 38 resolved, 14 overdue. Solid Waste: 25 open complaints (highest workload). Wards 6 and 7 have the highest complaint density."
Officer	"Which complaints appear to be related to the same underlying issue?"	"Complaint IDs: 1042, 1051, 1067. All refer to garbage at Tuljapur Naka, Ward 1. Group ID: G-12. Suggest merging into a single task."
Administrator	"Recommend resource allocation for next week."	"Solid Waste department is at 150% capacity in Ward 1. Recommend transferring 2 officers from Public Works to Solid Waste for Zone 1."

12. Feasibility & Time Commitment

Component	Complexity	Estimated Time
Citizen Portal (Mobile)	Medium	2 days
Field Worker App (Mobile)	Medium	1 day
Web Dashboard (Admin + Officer)	High	3 days
Image Classification (ResNet50)	Low (pre-trained)	0.5 day
Text Classification (TF-IDF)	Low	0.5 day
Duplicate Detection (Text + GPS)	Medium	1 day
Trend Detection (Anomaly)	Low (statistical)	0.5 day
Before/After Verification (SSIM)	Low (pixel math)	0.5 day
GenAI Assistant (RAG)	Medium	1 day
Solapur Data Seeding (Wards/Zones)	Low	0.5 day
SLA & Escalation Logic	Medium	1 day
Database + APIs	Medium	2 days
Total		~14 days (feasible for Buildathon)

13. The "Demo Day" Script (Winning Presentation)

Time	Segment	What to Show
0:00 - 0:30	The Problem	Show a flooded street in Solapur. <i>"In July 2026, 51 SMC officers were fined for delaying complaint resolution. Why? Because complaints get stuck—no auto-routing, no prioritization, no SLA tracking."</i>

Time	Segment	What to Show
0:30 - 1:30	Citizen Submission	Open app. Take a photo from Tuljapur Naka, Ward 1. Type <i>"Garbage piled up for 4 days."</i> Submit. Show auto-detected category (Garbage), severity (High), department (Solid Waste), and ward (Ward 1).
1:30 - 2:30	Duplicate Detection	Show two separate complaints (same pothole near City Mall, Ward 4) automatically grouped. <i>"We save officer time by merging duplicates."</i>
2:30 - 3:30	Officer Dashboard	Show the heatmap of Solapur with ward-level complaint density. Show emerging crisis alert: ■■■ <i>Garbage complaints in Ward 1 increased by 240%.</i> Show priority scoring.
3:30 - 4:30	Field Worker Resolution	Switch to field worker app. Show assignment, GPS tracking, resolution photo upload.
4:30 - 5:30	Before/After Verification	Slide a slider to compare before/after photos. <i>"Our system automatically verifies if the issue is actually fixed."</i>
5:30 - 6:30	SLA Auto-Enforcement	Show the SLA escalation workflow. <i>"If a complaint isn't assigned within 2 hours, the Department Head gets a reminder. After 4 hours, it escalates to the Municipal Administrator."</i>
6:30 - 7:00	GenAI Assistant	Ask <i>"What are the major problems in Ward 4?"</i> Show the assistant retrieving actual complaint data and generating a summary.
7:00 - 7:30	The Innovation	<i>"We don't just log complaints. We detect crises before they escalate, verify resolutions without manual inspections, and enforce SLAs to prevent the exact delays that got 51 SMC officers fined."</i>

14. Expected Deliverables (From PS2)

Deliverable	Status
Working Mobile Application (Citizen + Field Worker)	■ Planned
Working Web Application (Command Center + Operations)	■ Planned
Backend Services and APIs (FastAPI)	■ Planned
Persistent Database (PostgreSQL + PostGIS + pgvector)	■ Planned

Deliverable	Status
Database Schema Design	■ Provided
Role-based Authentication and Authorization (JWT)	■ Planned
AI/ML Implementation (ResNet50 + TF-IDF + SSIM)	■ Planned
GenAI Implementation (RAG + OpenAI)	■ Planned
Architecture Diagram	■ Textual description provided
API Documentation	■ Endpoints planned
README and Full Documentation	■ This document
Demonstration of Complete Complaint Lifecycle	■ Workflows documented
Final Presentation and Demonstration	■ Demo script provided

15. Out of Scope

To maintain the intended Buildathon scope, we are **not** implementing:

- Integration with actual municipal ERP systems
- Real government authentication systems
- Production-scale city-wide deployment
- Physical IoT sensors
- Autonomous field vehicles
- Real-time CCTV infrastructure
- Actual emergency dispatch infrastructure
- Payment systems
- Legal enforcement automation

Synthetic or organizer-provided operational records may be used where necessary.

End of Document